

Student:	Email:
Prakriti Dhungel	prakriti.dhungel98@gmail.com

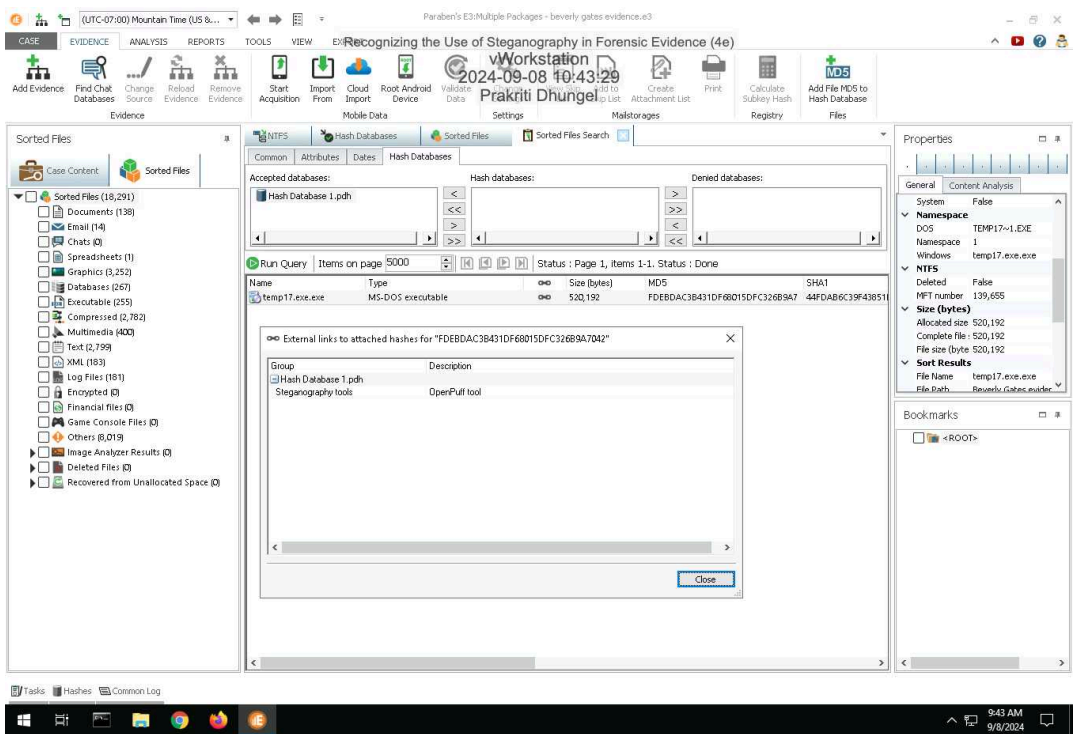
Time on Task:	Progress:
9 hours, 38 minutes	100%

Report Generated: Tuesday, September 17, 2024 at 1:23 PM

Section 1: Hands-On Demonstration

Part 1: Detect Steganography Software on a Drive Image

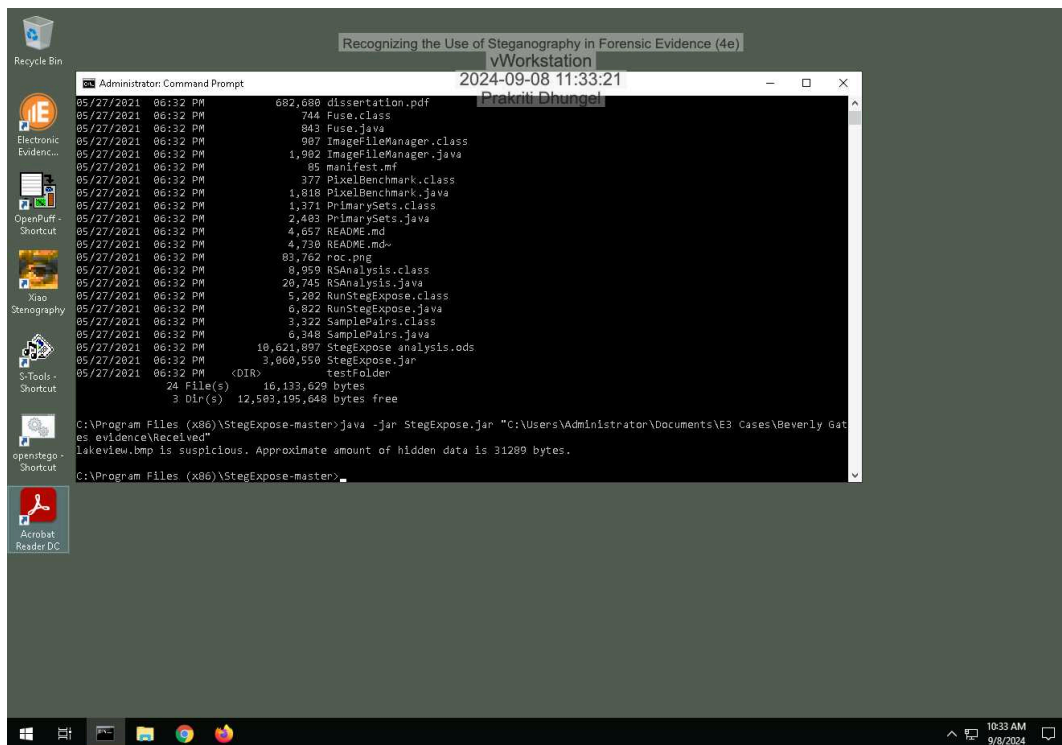
14. Make a screen capture showing the search result and its description.



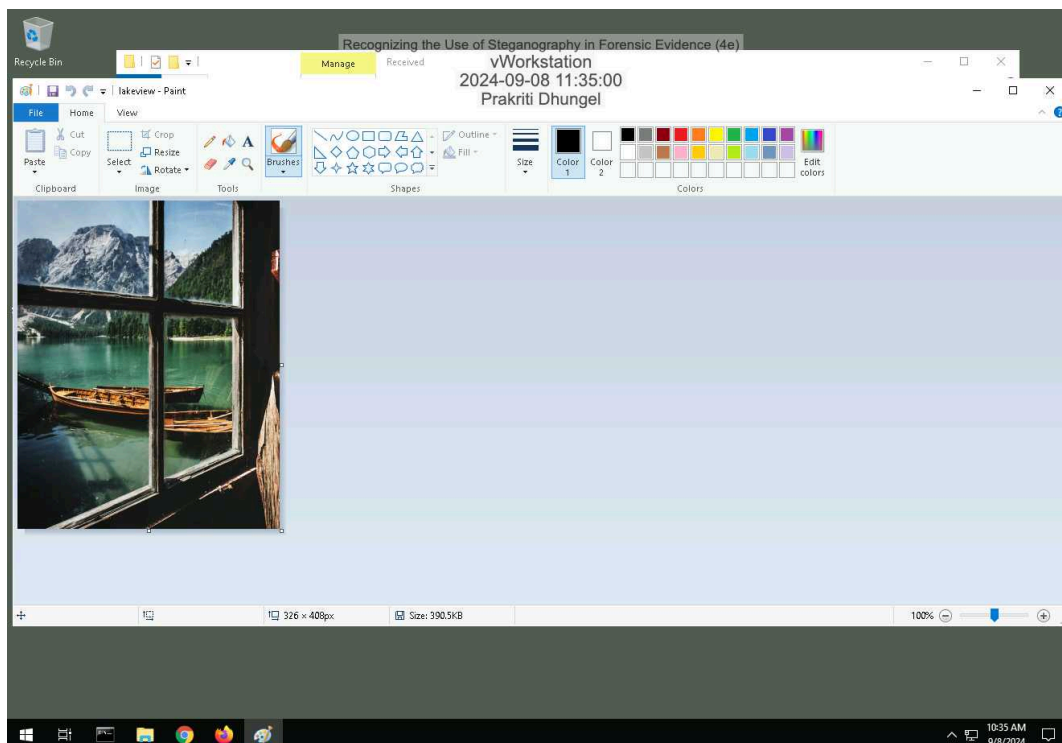
Hidden Open Puff Steganography tool inside temp17.exe.exe file.

Part 2: Detect Hidden Data in Image Files

10. Make a screen capture showing the StegExpose results.



13. Make a screen capture showing the suspicious file in Microsoft Paint.



Part 3: Extract Hidden Data from Image Files

2. Record the passphrase saved in the ReadMe file.

landmarks

16. Make a screen capture showing the contents of the file extracted by OpenPuff.



The screenshot shows a Notepad window titled 'locations - Notepad'. The text inside is a list of Chicago landmarks, each preceded by a coordinate pair (latitude, longitude) and followed by the address and city/state. The list includes landmarks such as '444 North Michigan', '55 E. Jackson', 'Altgeld & Lincoln', '670 N. Michigan', 'AT&T Building', 'Belmont & Leavitt', 'Beverly', 'Andersonville', 'Ashland & Altgeld', 'Brickyard-Diversey', 'Broadway & Clarendon', 'Broadway and Lawrence', 'Bucktown', 'Chicago & Franklin', 'Chicago & Wabash', 'Chicago Amoco Building', 'Clark & Belmont', 'Clark & Madison', 'Clybourn & Webster', 'Clark Street', 'Dearborn Park', 'Dearborn & Division', 'Diversey', 'Dickens', 'Division & Paulina', 'Edgewater', 'Embassy Suites', 'Germania Place', 'Hyde Park', 'East Wacker Drive', 'Irving Park', 'Edgewater', 'Hyde Park', 'Lake & LaSalle', 'LaSalle & Monroe', 'Leo Burnett', 'Lincoln & Greenview', 'Lincoln & Belden', 'Lincoln & Wilson', 'Lincoln & Paulina', 'Logan Blvd.', 'Lincoln Park-Fullerton', and 'Merchandise Mart'.

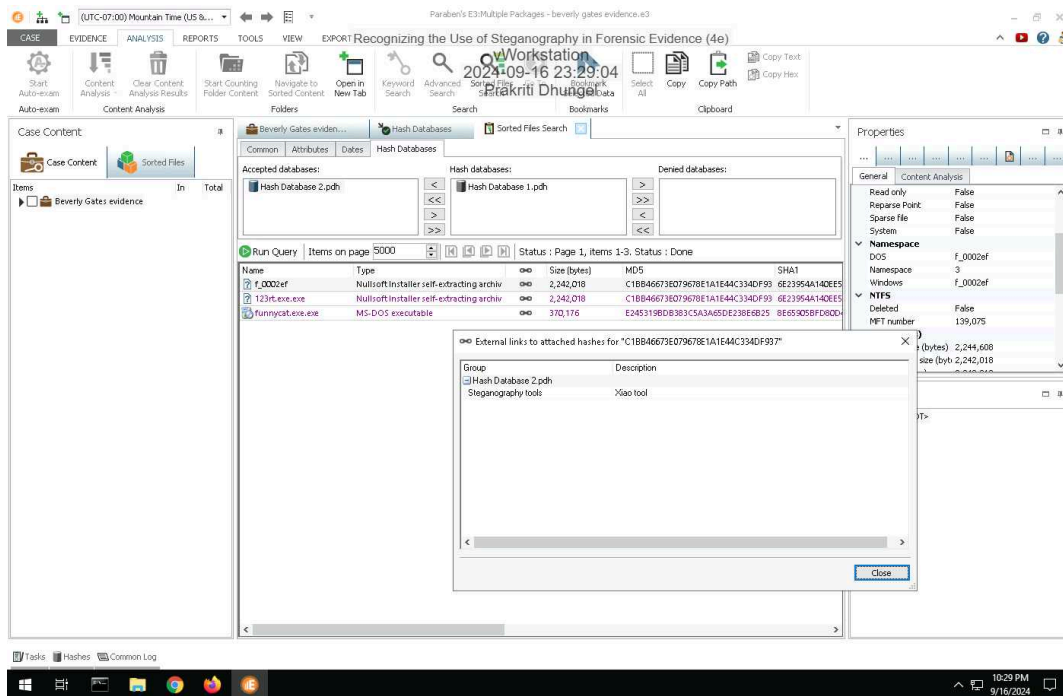
17. Describe the contents of the hidden file. How might it be relevant to the current investigation?

The hidden file contains the GPS coordinates, longitude and latitude of Chicago Illinois along with the name of the buildings. This might be the data of people who might be involved in drug trafficking with Beverly Gates. They could be the customers or the suppliers of her.

Section 2: Applied Learning

Part 1: Detect Steganography Software on a Drive Image

5. Make a screen capture showing the search result and its description.



Part 2: Detect Hidden Data in Image and Audio Files

4. Identify the image file with concealed data according to the StegExpose steganalysis tool.

According to the StegExpose tool, db9olser.gif is suspicious.

Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

7. Make a screen capture showing the WAV file sizes and hash values in E3.

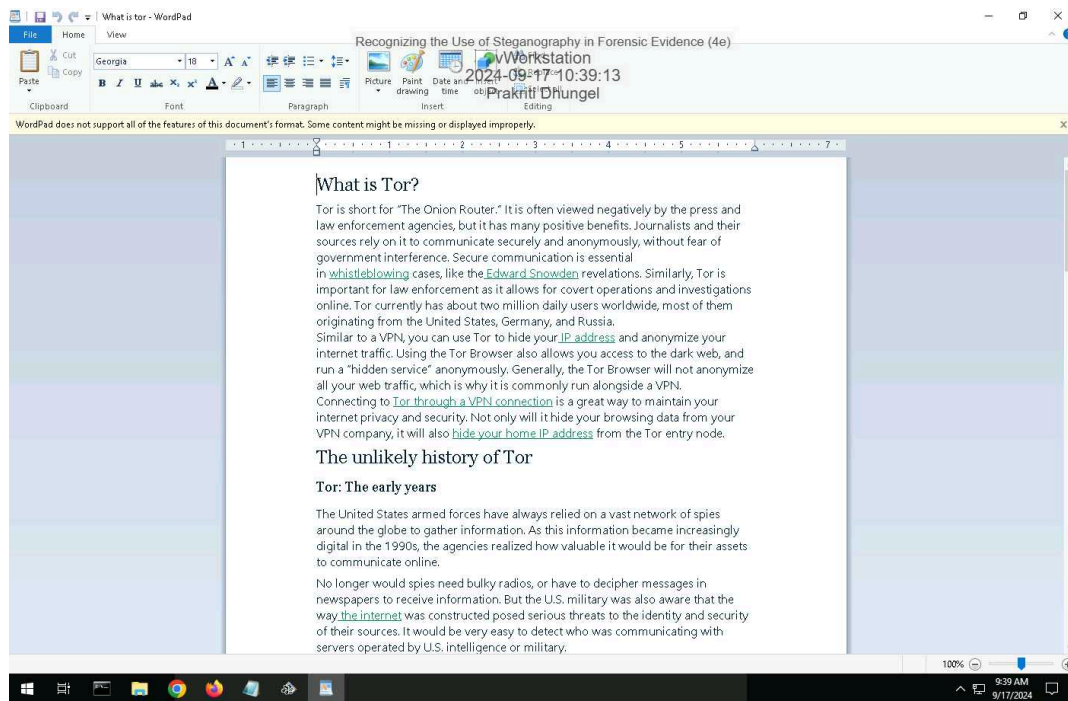
The screenshot displays the Paraben's E3 software interface. The main window shows a list of files under the 'Internal Path: e3://Sorted Files/Multimedia' directory. The files are sorted by name, and the columns include Name, Type, Size (bytes), MD5, and SHA1. The files listed are:

Name	Type	Size (bytes)	MD5	SHA1
animal_tick.mp3	Audio file with ID3 version 2	13,693	74D2956CE9C363E4AD0F08049576C	26E91DE390
h32.mp3	Audio file with ID3 version 2	36,937	93A8F8C9F5C736D23A4FEB860C38D	37F059A0D1
beep_tree.mp3	Audio file with ID3 version 2	16,093	D819AC371F54236ACCE5A86C0B52	97F0B0A44D
calls_alert_v2.mp3	Audio file with ID3 version 2	9,656	5C2E5A7173C74AEEF17D9CA8BE09F	CEFD04FB4C
calls_confirmation_v2.mp3	Audio file with ID3 version 2	8,611	D63AAE8B30B71F1A20F0AD7FCAA8	10267FCE7E
calls_incoming_ring_v2.m	Audio file with ID3 version 2	37,450	C3CC0C45D1A49A54B6B0D3E76B	7A1A19025D
calls_outgoing_ring_v2.m	Audio file with ID3 version 2	17,388	11D8EB312D50D801058139CF8D86	4388CFD069
calls_pop_v2.mp3	Audio file with ID3 version 2	5,267	D2B79FAC48FCE5BAED00F03C0DF44	E88115404E
calls_they_joined_call_v	Audio file with ID3 version 2	9,656	1A6F3906B8A91BBE27B83ACB51867	A3E550FDE2
calls_they_left_call_v2.m	Audio file with ID3 version 2	9,656	451658EC5BAF2623825A407A80957	63413C0E09
calls_you_joined_call_v2	Audio file with ID3 version 2	11,954	AD7F0074D0C7E6792385850B87D0C2	D99C46D74D
calls_you_left_call_v2.m	Audio file with ID3 version 2	11,954	F9A8F9AC426A9E4A25F7F551D0	D2B8F30C2E
complete_quest_require	Audio file with ID3 version 2	30,576	BF1D81A7A442C4D95F7D05A4D198FC5	7C0E48B36
confirm_delivery.mp3	Audio file with ID3 version 2	20,695	CC9D0F11D68F82ED163402C1D1BCB	663F3C876
fitterbug.mp3	Audio file with ID3 version 2	44,462	7D3D8D43C9CD16D3761C878413A3F	5300CD87E
here_you_go_lighter.mp3	Audio file with ID3 version 2	17,157	3C1F056325C2362454615FD1C2D12	B8E1493211
hi_flower_hut.mp3	Audio file with ID3 version 2	23,878	A44D05872D5C1158BED679418E9721	42E89738F1
hummus.mp3	Audio file with ID3 version 2	13,270	20DE354C96D5901F42339A64167158C	32D0548D4E
item_pickup.mp3	Audio file with ID3 version 2	12,455	9EAC37F227789E7ED69695020E633	F7C70DF012
knock_brush.mp3	Audio file with ID3 version 2	15,067	8D6E1624815AC656FE37A4B213918	B5CA41BB11
save_and_checkout.mp3	Audio file with ID3 version 2	37,400	1F16C1A6C0798C1C2D29F2828C457	25C2EF8526
MANIFEST-000002	MPEG-4 LOAS	50	228F08B1658B16452518158748B3D14	5695502035
F_000010	WMV4 video file	44,980	0B0CD09F53A1366F8B1E5D1514163	A3310C2093
F_000022	Audio file with ID3 version 2	19,808	AETD16B83EEA71B98997D06FAD666	AC208E3969
F_000027	Audio file with ID3 version 2	17,922	DD93D0C6A01E588B80676581E2905	AAAA471151
motorcycle_x.wav	Microsoft WAV file	53,247	47A41CF04BC9F20C0D76A5B64990C	8726CEBF11
motorcycle_x1.wav	Microsoft WAV file	53,248	F93E98018308D719A3C7B8B8407421	83DAC4C825
laugh_x.wav	Microsoft WAV file	14,204	35C3A81D30A31F8BF70B23687B3390	A8F6A8A4D
laugh_x1.wav	Microsoft WAV file	14,205	5D5289FEA9E4546A15F06FDDA5E0	8DCCD064E

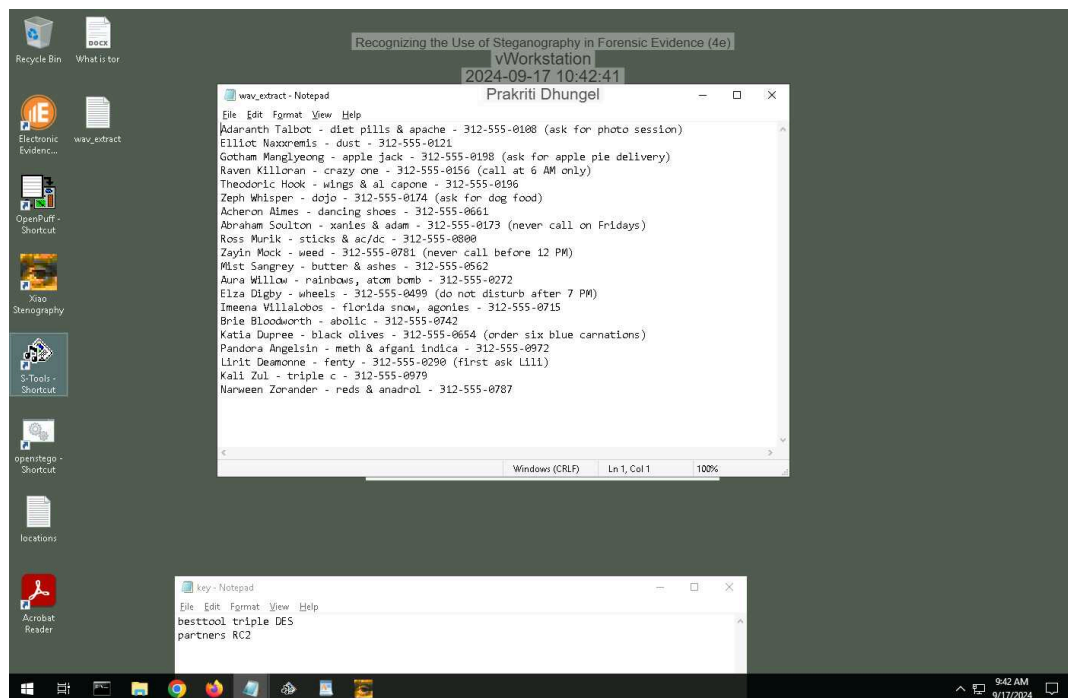
The right-hand pane shows the 'Properties' window for the selected file, displaying details such as 'Read only', 'Repair Point', 'Sparse file', 'System', 'Namespace', 'MFT number', and 'Size (bytes)'. The 'Size (bytes)' section indicates an allocated size of 16,394 bytes and a complete file size of 14,204 bytes.

Part 3: Extract Hidden Data from Image and Audio Files

9. Make a screen capture showing the contents of the hidden file extracted by S-Tools.



15. Make a screen capture showing the contents of the hidden file extracted by Xiao.



Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

16. **Describe** the contents of the two hidden files. How might they be relevant to the current investigation?

laugh_x1.wav seems to be a suspicious modified file with hidden file in it that contains list of drugs names and their supplier's information that are relevant to it. Whereas laugh_x.wav seems to be an original file.

Section 3: Challenge and Analysis

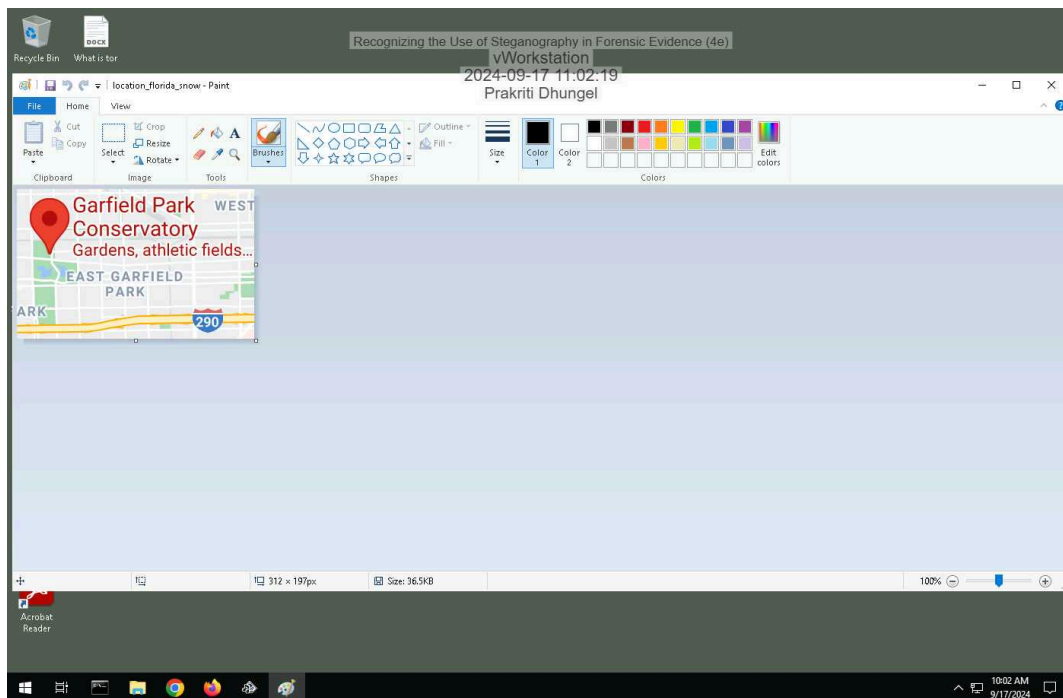
Part 1: Detect More Hidden Data

Record the names of the files that contain concealed data.

chicago.bmp is suspicious. Approximate amount of hidden data is 114062 bytes. chicago1.bmp is suspicious =. Approximate amount of hidden data is 57942 bytes.

Part 2: Extract More Hidden Data

Make a screen capture showing the **first file extracted by OpenStego**.



Recognizing the Use of Steganography in Forensic Evidence (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 02

Make a screen capture showing the second file extracted by OpenStego.

